

Erdős Problem #323

EP-323

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $1 \leq m \leq k$ and $f_{k,m}(x)$ denote the number of integers $\leq x$ which are the sum of m many nonnegative k th powers. Is it true that $f_{k,k}(x) \gg_{\epsilon} x^{1-\epsilon}$ for all $\epsilon > 0$? Is it true that if $m < k$ then $f_{k,m}(x) \gg x^{m/k}$ for sufficiently large x ?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>